Equazioni dell'osservatore adattativo (7)

Le equazioni dell'osservatore in assi statorici d–q sono le seguenti:

$$\begin{cases} L_\mathrm{s}\dfrac{\mathrm{d}}{\mathrm{d}t}\hat{i}_\mathrm{sd} = -R_\mathrm{s}\hat{i}_\mathrm{sd} + (\omega_0 - p\hat{\omega})\, L_\mathrm{s} i_\mathrm{sq} + p\hat{\omega}\hat{\psi}_\mathrm{sq} + u_\mathrm{sd} + \eta_\mathrm{d} \\ L_\mathrm{s}\dfrac{\mathrm{d}}{\mathrm{d}t}\hat{i}_\mathrm{sq} = -R_\mathrm{s}\hat{i}_\mathrm{sq} - (\omega_0 - p\hat{\omega})\, L_\mathrm{s} i_\mathrm{sd} - p\hat{\omega}\hat{\psi}_\mathrm{sd} + u_\mathrm{sq} + \eta_\mathrm{q} \\ \dfrac{\mathrm{d}}{\mathrm{d}t}\hat{\psi}_\mathrm{sd} = -R_\mathrm{s} i_\mathrm{sd} + \omega_0\hat{\psi}_\mathrm{sq} + u_\mathrm{sd} + \xi_\mathrm{sd} \\ \dfrac{\mathrm{d}}{\mathrm{d}t}\hat{\psi}_\mathrm{sq} = -R_\mathrm{s} i_\mathrm{sq} - \omega_0\hat{\psi}_\mathrm{sd} + u_\mathrm{sq} + \xi_\mathrm{sq} \\ \dfrac{\mathrm{d}}{\mathrm{d}t}\hat{\chi}_\mathrm{d} = \omega_0\hat{\chi}_\mathrm{q} - k_\mathrm{i}\tilde{i}_\mathrm{sq} \\ \dfrac{\mathrm{d}}{\mathrm{d}t}\hat{\chi}_\mathrm{q} = -\omega_0\hat{\chi}_\mathrm{d} + k_\mathrm{i}\tilde{i}_\mathrm{sd} \\ \dfrac{\mathrm{d}}{\mathrm{d}t}\hat{\omega} = -\dfrac{1}{\gamma}\big[(\hat{\psi}_\mathrm{sq} - L_\mathrm{s}\hat{i}_\mathrm{sq})\tilde{i}_\mathrm{sd} - (\hat{\psi}_\mathrm{sd} - L_\mathrm{s}\hat{i}_\mathrm{sd})\tilde{i}_\mathrm{sq}\big] \\ \eta_\mathrm{d} = k_\mathrm{p}\tilde{i}_\mathrm{sd} + p\hat{\chi}_\mathrm{q} \\ \eta_\mathrm{q} = k_\mathrm{p}\tilde{i}_\mathrm{sq} - p\hat{\chi}_\mathrm{d} \\ \xi_\mathrm{sd} = -k_\psi(\hat{\psi}_\mathrm{sd} - \psi^*_\mathrm{sd}) \\ \xi_\mathrm{sq} = -k_\psi(\hat{\psi}_\mathrm{sq} - \psi^*_\mathrm{sq}) \\ \psi^*_\mathrm{sd} = \varphi_\mathrm{m} + L_\mathrm{s} i^*_\mathrm{sd} \\ \psi^*_\mathrm{sq} = L_\mathrm{s} i^*_\mathrm{sq} \end{cases} \tag{0.1}$$

dove il carattere $\hat{}$ indica le grandezze stimate e il carattere $\tilde{}$ indica l'errore di stima della relativa variabile (ad esempio $\tilde{i}_{sd} = i_{sd} - \hat{i}_{sd}$).

Le variabili ausiliarie η_d, η_q, ε_{sd}, ε_{sq} sono definite per garantire la stabilità dell'osservatore tramite tecniche di Lyapunov e per determinarne le prestazioni.

$$\begin{cases} L_\mathrm{s}\dfrac{\mathrm{d}}{\mathrm{d}t}\hat{i}_\mathrm{sd} = -R_\mathrm{s}\hat{i}_\mathrm{sd} + (\omega_0 - \hat{\omega})\, L_\mathrm{s} i_\mathrm{sq} + p\hat{\omega}\hat{\psi}_\mathrm{sq} + u_\mathrm{sd} + \eta_\mathrm{d} \\ L_\mathrm{s}\dfrac{\mathrm{d}}{\mathrm{d}t}\hat{i}_\mathrm{sq} = -R_\mathrm{s}\hat{i}_\mathrm{sq} - (\omega_0 - \hat{\omega})\, L_\mathrm{s} i_\mathrm{sd} - \hat{\omega}\hat{\psi}_\mathrm{sd} + u_\mathrm{sq} + \eta_\mathrm{q} \\ \dfrac{\mathrm{d}}{\mathrm{d}t}\hat{\psi}_\mathrm{sd} = -R_\mathrm{s} i_\mathrm{sd} + \omega_0\hat{\psi}_\mathrm{sq} + u_\mathrm{sd} + \xi_\mathrm{sd} \\ \dfrac{\mathrm{d}}{\mathrm{d}t}\hat{\psi}_\mathrm{sq} = -R_\mathrm{s} i_\mathrm{sq} - \omega_0\hat{\psi}_\mathrm{sd} + u_\mathrm{sq} + \xi_\mathrm{sq} \\ \dfrac{\mathrm{d}}{\mathrm{d}t}\hat{\omega} = -\dfrac{1}{\gamma}\big[(\hat{\psi}_\mathrm{sq} - L_\mathrm{s}\hat{i}_\mathrm{sq})\tilde{i}_\mathrm{sd} - (\hat{\psi}_\mathrm{sd} - L_\mathrm{s}\hat{i}_\mathrm{sd})\tilde{i}_\mathrm{sq}\big] \end{cases} \tag{0.2}$$

$$
\begin{cases}
\eta_{\mathrm{d}} = k_{\mathrm{p}}\tilde{i}_{\mathrm{sd}} + \hat{\chi}_{\mathrm{q}} \\
\eta_{\mathrm{q}} = k_{\mathrm{p}}\tilde{i}_{\mathrm{sq}} - \hat{\chi}_{\mathrm{d}} \\
\dfrac{\mathrm{d}}{\mathrm{d}t}\hat{\chi}_{\mathrm{d}} = \omega_0\hat{\chi}_{\mathrm{q}} - k_{\mathrm{i}}\tilde{i}_{\mathrm{sq}} \\
\dfrac{\mathrm{d}}{\mathrm{d}t}\hat{\chi}_{\mathrm{q}} = -\omega_0\hat{\chi}_{\mathrm{d}} + k_{\mathrm{i}}\tilde{i}_{\mathrm{sd}} \\
\xi_{\mathrm{sd}} = -k_{\psi}\left(\hat{\psi}_{\mathrm{sd}} - \psi^*_{\mathrm{sd}}\right) \\
\xi_{\mathrm{sq}} = -k_{\psi}\left(\hat{\psi}_{\mathrm{sq}} - \psi^*_{\mathrm{sq}}\right) \\
\psi^*_{\mathrm{sd}} = \varphi_{\mathrm{m}} + L_{\mathrm{s}}i^*_{\mathrm{sd}} \\
\psi^*_{\mathrm{sq}} = L_{\mathrm{s}}i^*_{\mathrm{sq}}
\end{cases}
\tag{0.3}
$$